Exploring limitations of BERT for End-to-End Aspect-based Sentiment Analysis

Tabbatha Motley
https://github.com/SynthNibbler/BERT-E2E-ABSA

1. Introduction

Communicative language is complex and nuance. Each domain and field has its jargon and unique connotations. This creates a challenge for AI models with natural language processing to accurately interpret human prose and predict the best responses. "Sentiment analysis (SA) is an intellectual process of extricating user's feelings and emotions" [1]. Models can use this technique on different scales; document level sentiment analysis determines the polarity of the entire document. A single sentiment is determined for the entire body of text. While sentence-level determines a single sentiment for the whole sentence. Conversational language is rarely simple enough to determine a singular sentiment. Aspect-based sentiment analysis (ABSA) is the fine-grain method of determining polarity of target tokens, or aspects. . A sentence can have zero to many aspects within it. In the research of Xin Li et al "Exploiting BERT for End-to-End Aspect-based Sentiment Analysis" [2] several sizes of the BERT model are explored to use for the ABSA task, specifically E2E-ABSA, end-to-end aspect-based sentiment analysis. For E2E-ABSA a target aspect will have a linked opinion term that determines the sentiment of that aspect within a sentence. Both the aspect and the opinion term must be identified by the model to determine the aspect polarity. This research aims to show that pre-trained models offer fast and promising results for ABSA research. The f1 benchmark is shown to be close and even exceeding the state-of-the-art models of the time that where build specially for this task. While f1 benchmarks do show an aggregate representation of the performance of a model, it does reveal anything about its errors or limitations. What did the metrics hide in BERTs performance in the ABSA task of E2E, especially what weak points are not obvious from the aggregated metrics? This project replicates the original BERT+Linear from this work and probes different subsets of data to reveal the challenges and limitations.

2. Background and related work

"Aspect-based sentiment analysis (ABSA) is the problem to identify sentiment elements of interest for a concerned text item,either a single sentiment element,or multiple elements with the dependency relation between them" [3]. ABSA is a task that targets an entity within a sentence and determines the sentiment for each by extracting the pairs, aspect and opinion term. Using contextual evidence the analysis will determine the polarity of the aspect based on the opinion term and other context depending the design of the down stream task. Many common benchmark datasets are listed in the paper "A

Survey on Aspect-Based Sentiment Analysis: Tasks, Methods, and Challenges" [3]. This paper also introduces several model paradigms used for different ABSA tasks, pre-trained models are noted to have a rising frequency in research use [3].

The research by Xin Li et al "Exploiting BERT for End-to-End Aspect-based Sentiment Analysis" claims BERT + Linear has good performance on E2E tasks even though it was the simplest BERT model used [2]. End-to-End ABSA is a compound ABSA task that uses a pair extraction to determine both aspect and opinion term. This method focuses on finding aspect terms and their matching sentiment polarity within sentences simultaneously. This paper claims that using BERT for end-to-end ABSA reveals a pre-trained model can perform ABSA tasks on benchmark datasets, SemEval-2014 [4] and SemEval-2015 [5], with metrics close to those of specially trained models built specifically for the E2E-ABSA task [2].

This was achieved by training a task specific layer over the base model that mapped the standard BERT output to a ABSA task labeling scheme. This linear layer is considered a fine-tuning of the model as this layer is trained on the benchmark dataset Laptop-14 from the SemEval-2014 and SemEval-2015 altered by X. Li [6]. This research used two methods for the fine-tuning process. Frozen BERT method chose to fine-tune by training the model and freezing the base models weights. This keeps the weights the model learned through the pre-training process and only trains the linear layer for the ABSA task on the datasets. The other method was noted to be more successful, unfrozen BERT. This method allows all weights to be adjusted during the training with the datasets [2]. The paper claims this success is due to the diverse domains a pre-trained model is trained on giving it a range of contextual knowledge making it a strong choice for E2E-ABSA task since it only needs a fast fine-tuning to hit a competitive F1 performance.

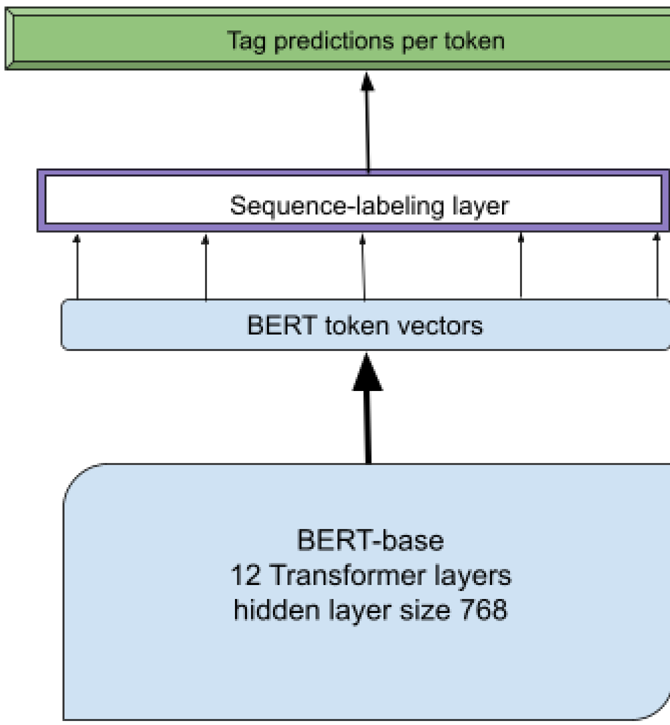
The experiments of this research focused on detecting the aspect within the sentence and the sentiment of the opinion term it was paired with. The model only output the target aspect and its polarity. The metrics do not calculate accurately the category prediction of the aspect nor does it calculate accurate identification of the opinion term. The f1 score is only calculated from the predicted aspect and polarity when compared to the gold labels, or ground truth found in the annotations of the dataset.

The paper does not elaborate on errors or where the model struggled most that effected its f1 score. Instead they focus on the model not overfitting and assert the pre-trained model with a fine-tuning process that is fast to replicate allowed research on ABSA without needed a complex custom built model for the task. This motivates the question about what is hiding in the aggregated metrics from this paper. To experiment with this fine-tuned model the experiment would need to be replicated so that the question can be explored.

3. Model Architecture Replicated

The replication used BERT+Linear to explore the simplest model since the research emphasises its performance as interesting due to its small size. The base model used was bert-base-uncased with hidden layer size dim_h = 768, 12 transformer layers with 12 attention heads and a feed-forward size of 3072. The number of trainable parameters is 110 million [7]. An additional layer, the Linear layer or the sequence-labeling layer, is used as a task specific layer to map the BERT token vectors to ABSA labeling, (POS, NEG, NEU). Finally the model outputs the predictions per token by tagging. Example output ['pizza']:POS for the aspect "pizza" with a positive polarity.

Figure 1. Fine-tuned layer and base model



4. Data Details

The dataset used to finetune the model was laptop-14 from SemEval 2014 after being altered by X. Li et al in research for aspect target extraction [2]. The work with BERT by X. Li used the Laptop domain dataset with 2041 aspects in the training set, 256 aspects for the validation set and 634 aspects for the hold out test set used to product the aggregate metrics [2]. This same data split was used for the replication.

Laptop-14:	Train	Validation	Test
Aspects	2041	256	634

The dataset used in replication is the altered version of the SemEval 2014 Laptop-domain set by X. Li et al and it is not stated in either paper how the classes are split. According to M. Pontiki et al in “SemEval-2014 Task 4: Aspect based sentiment analysis” and “SemEval-2015 Task 12: Aspect based sentiment analysis” where the datasets where originally annotated, the class "POS" had a large dominance in the set [4]. There was also a class "CON" for conflict in a sentence caused by a discourse term such as "but," "although," or comparison terms. These terms were tagged as "CON" but made up a very small class compared to POS/NEG/NEU [4]. Laptop Train had 45 conflict aspects with a total of 2358 making this class less than 2% of the training set classes.

Dataset	Pos.	Neg.	Con.	Neu.	Tot.
LPT-TR	987	866	45	460	2358
LPT-TE	341	128	16	169	654

The experiment with BERT only used a training set of 2041 aspects of unknown class counts. However, inspecting the raw datasets shows that conflict terms and sentiment is still present in the dataset just not explicitly labeled [7]. The class for conflict was not used the in the experiment with BERT by X. Li et al. Only the classes POS/NEG/NEU were used so it is unclear which class dominates the set used and how many conflict rich sentences were kept.

5. Methodology for replication

To properly replicate the experiment for BERT the environment was set up using a csuchico server with access to an A100 gpu. The original repository was cloned down once the environment was set and activated. The requirements listed in "requirements.txt" [7] are:

numpy==1.22.0
tensorboardX==1.8
torch==1.2.0
tqdm==4.32.1
transformers==4.1.1

The replication achieved:

numpy==1.16.4
tensorboardX==1.9
torch==1.5.1
tqdm==4.32.1
transformers==4.1.1

Discrepancies in versions for numpy, torch, and tensorboardX are due to compatibility issues with the A100 and the age of the original repository. Some availability of older downloads was also limited. The original requirements were followed as closely as possible while assuring the work could be done on the csuchico server and have access to the A100 gpu. Once the environment was as close to the requirements as possible the environment was frozen to prevent any accidental or automatic updates. The "readme.md" provided basic instructions for how to replicate the experiment done in the paper for BERT+Linear using a script "fast_run.py" that trains the model on five different seeds and selects the best checkpoint [7]. Each training run completes 1500 epochs with a learning rate of 2e-5 [2]. This created an experimental baseline to probe the model with curated tests to see where the model was strongest and where it struggled. A common limitation of even modern models is around discourse terms, compare and contrasting sentence structures and sarcasm [8]. Discourse terms were selected as a probe to test the model on these more challenging sentence structures that show up in natural language.

The model was successfully fine-tuned to the E2E-ABSA task on the dataset laptop-14 and baseline very close to the original experiment was achieved. The replication results are a slightly lower precision score and a slightly higher recall rate, with a micro-f1 score slightly higher overall. The best replication training cycle presented an f1 score of 60.68 while the work by X. Li produced and f1 score of 60.43 [2], a difference of only 0.25.

Replication:	precision	recall	f1
BERT+Linear	61.63	59.78	60.68

	precision	recall	f1
BERT+Linear	62.16	58.90	60.43

Benchmark results from [cite BERT-E2E]

6. Evaluating conflict and non-conflict

Subsets of Laptop-14 Test [7] were made by carving the file and cleaning it by hand to verify sentences were complete. The pure conflict subset was carved using keyterm searching for discourse terms such as "but", "however", "though" and comparisons. The pure non-conflict subset was a collection of randomly selected simple sentences that did not have any discourse terms or complex comparisons. Each is subset was used as a probe of 22-23 aspects carved as a subset from the original test set. The sample size was kept relatively small since so few discourse terms appear in the Laptop-14 test set. This created a limitation for each set, based on the pure conflict subset size. By using the carved Laptop-14 test set for the subsets, the metrics could be calculated on these subsets and directly compared to the benchmark achieved in the replication. This revealed a challenge for the model on pure conflict sentences.

7. Results and Interpretation

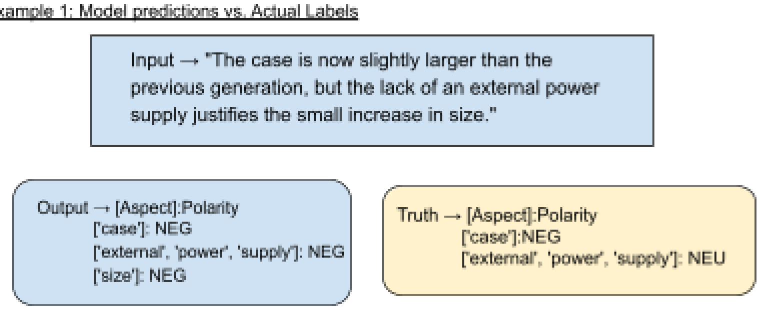
The results imply the model does perform better on sentence that lack discourse. However, it also reveals the f1 score hides this limitation. If the model could improve its performance on the pure conflict type sentences it would improve the model's over all performance. The spread from pure conflict to non-conflict f1 score is 23. This is a substantial gap hidden by the aggregate metrics presented in the original research for BERT. With a 14 point gap from the benchmark of the replication at 60.68 to the non-conflict subset, this gap is larger than from benchmark down to pure-conflict subset performance. Given the very small amount of conflict structured sentences present in the training set, the model's performance does not suffer too greatly on pure-conflict.

Test Type	Precision	Recall	micro-f1
*Full test set	61.63	59.78	60.68
non-conflict	78.95	71.43	74.99
conflict	48.15	56.52	51.99

8. Output and Error Analysis

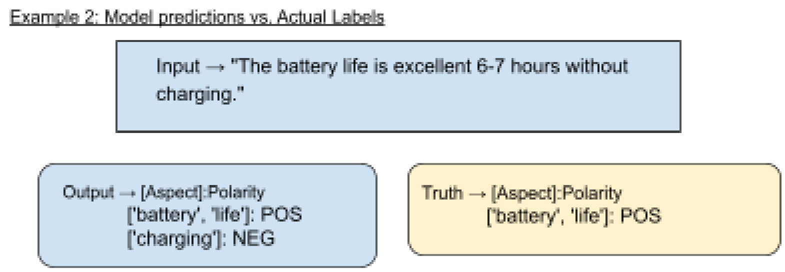
In the pure conflict subset:

Example 1: Model predictions vs. Actual Labels



In Example 1 the model predicts too many aspects compared to the gold labels, or truth. It does predict the full aspect with beginning inside and ending terms. However, it outputs a sentiment of negative, NEG while the truth is actually neutral, NEU. With domain context the "lack of an external power supply" for a laptop enhances its portability. A larger case hinders portability. The model's error on the sentiment the external power supply seems to ignore a domain nuance. It also selects an extra aspect "size" with a negative polarity. This prediction of the target aspect "size" seems like a reasonable prediction to a human reader, however it does not agree with the labeling as a ground truth. Therefore this is marked as an error, even though it might seem like a reasonable choice for aspect and polarity. If the case is what has increased in size then both must be negative. The prediction of case and a negative polarity is the only fully accurate prediction here.

Pure non-conflict subset:



In Example 2 we see a very simple sentence. Grammatically it may read a bit awkward to a human reader, but the sentiment is very straightforward; the battery life is excellent. The model correctly predicts the full target aspect "battery life" and its polarity as positive. However, the model makes an extra prediction of "charging" with a polarity as negative. This error could be due to the odd syntax in this sentence. While the model does perform better on the non-conflict subset, it also make similar mistakes by predicting extra aspects.

The error patterns the model makes in its predictions are similar and not distinct to one subset or the other. However, it makes far more errors on the pure-conflict set. It predicts too many aspects or attaches the wrong sentiment effecting precision. Occasionally it will miss an aspect all together effecting recall. The pure-conflict sentences have a more complex structure and tend to have more grammatically errors or are written as people speak instead of formal writing. These sentences could be difficult for even a human reader to determine correctly. Upon a closer look some labels seem more conservative about target aspects such as example 2 not labeling "size" as a true aspect, when it seems it would be an important element.

9. Conclusions and Future work

Aggregated metrics hide nuanced behaviors of the model such as limitations and strengths. To fully understand how to improve them the errors must be isolated and explored to be revealed. This way the improvements can target the failure patterns directly. Looking at performance alone presented in “Exploiting BERT for end-to-end aspect-based sentiment analysis” by X. Li et al the claims that pre-trained models outperform more complex models seem possible. However, the errors of the models needs to be explored and probed to see if they over come the challenges that exist in natural language processing methods [9].

Future work can explore the errors of other models and compared to BERT+Linear error patterns. Analyzing the limitations of the models beyond metrics can reveal the exact complexities in natural language that lead to common error patterns. Once the complexity is understood, solutions to improve models would be universal. Dataset annotation methods may also adjust how data is labeled to either explicitly accommodate conflict or allow enough context to be extracted to detect the dependencies implicitly.

10. Acknowledgements:

I would like to thank Dr. Bo Shen for his guidance and feedback throughout this project, especially with helping shape the direction of the conflict focused testing.

11. References

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AI disclosure

ChatGPT was collaborated with for troubleshooting the environment and comparing requirements to assure alignment. It was also used to generate IEEE citations of the sources used for reference.